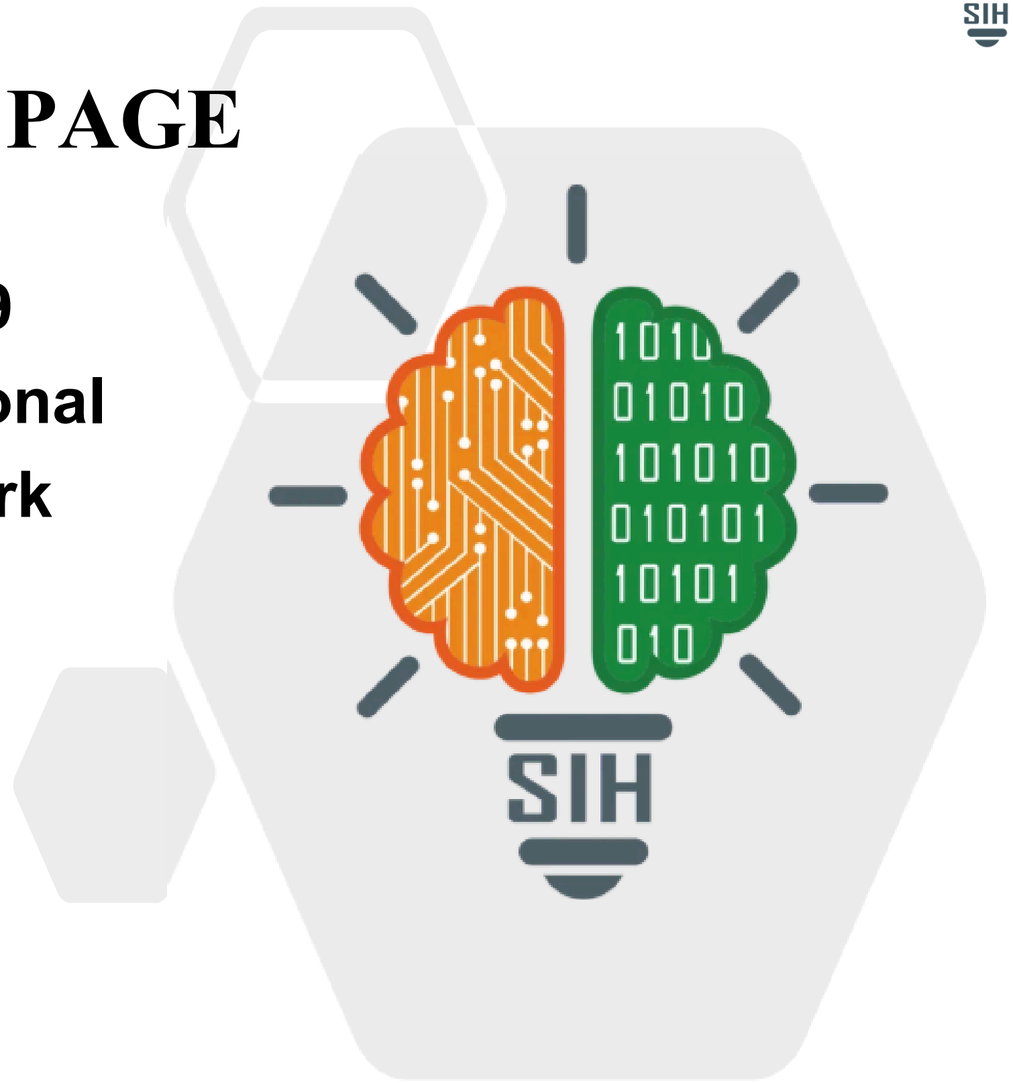
TITLE PAGE



- **Problem Statement ID – SIH26099**
- **Problem Statement Title - National Unified Material Master Framework**
- **Theme - Smart Automation**
- **PS Category- Software**
- **Team ID- 140981**
- **Team Name - SLUG**

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IDEA TITLE

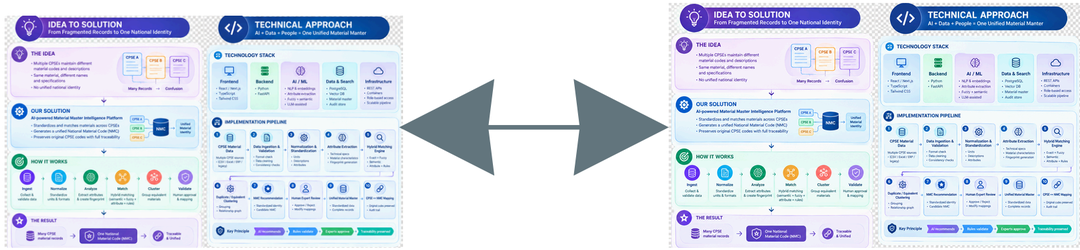
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PROBLEM:

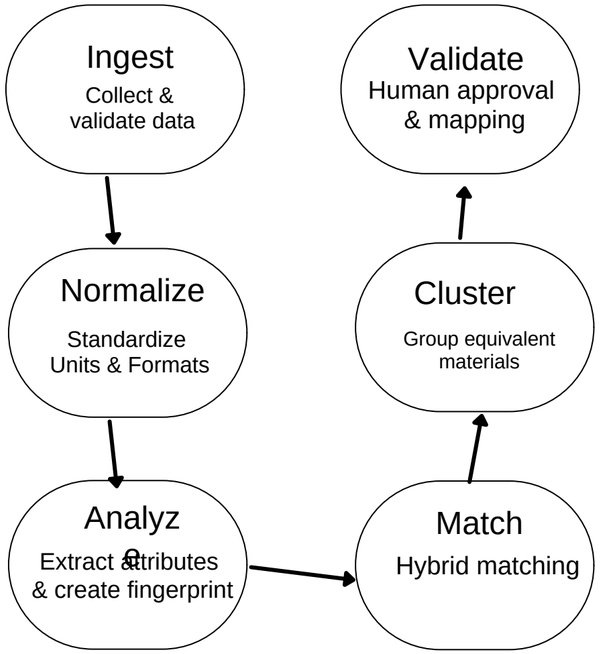
- **Fragmented Material Masters:** Different CPSEs use different codes, descriptions, units, specifications, and naming conventions for the same material.
- **Duplicate & Near-Duplicate Records:** Equivalent materials remain scattered across organizations, creating fragmented procurement information and higher inventory levels.
- **Technical Matching Risk:** Simple text matching can falsely merge incompatible materials such as SS304 vs SS316 or M16×50 vs M16×100.
- **Limited Traceability:** Standardization requires preserving legacy codes, source lineage, approval history, and the reasoning behind every AI recommendation.

SOLUTION - One Nation, One Material Code:

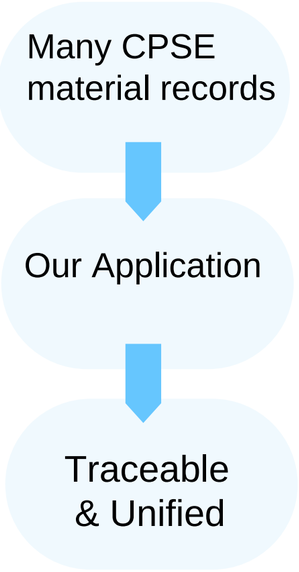
- **From fragmented records to one identity:** AI normalizes material data, extracts technical attributes, and generates structured material fingerprints.
- **From text similarity to intelligent matching:** Hybrid matching combines exact, fuzzy, semantic, attribute, and technical compatibility signals.
- **From duplicates to a National Master:** Equivalent materials are clustered and mapped to a persistent National Material Code (NMC) while preserving legacy CPSE codes.
- **From black-box AI to governed decisions:** Every recommendation provides matching evidence, confidence, conflicts, and source lineage for human approval.



How It Works



The Result



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TECHNICAL APPROACH

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Technology Stack

Frontend: React / Next.js, TypeScript , Tailwind CSS

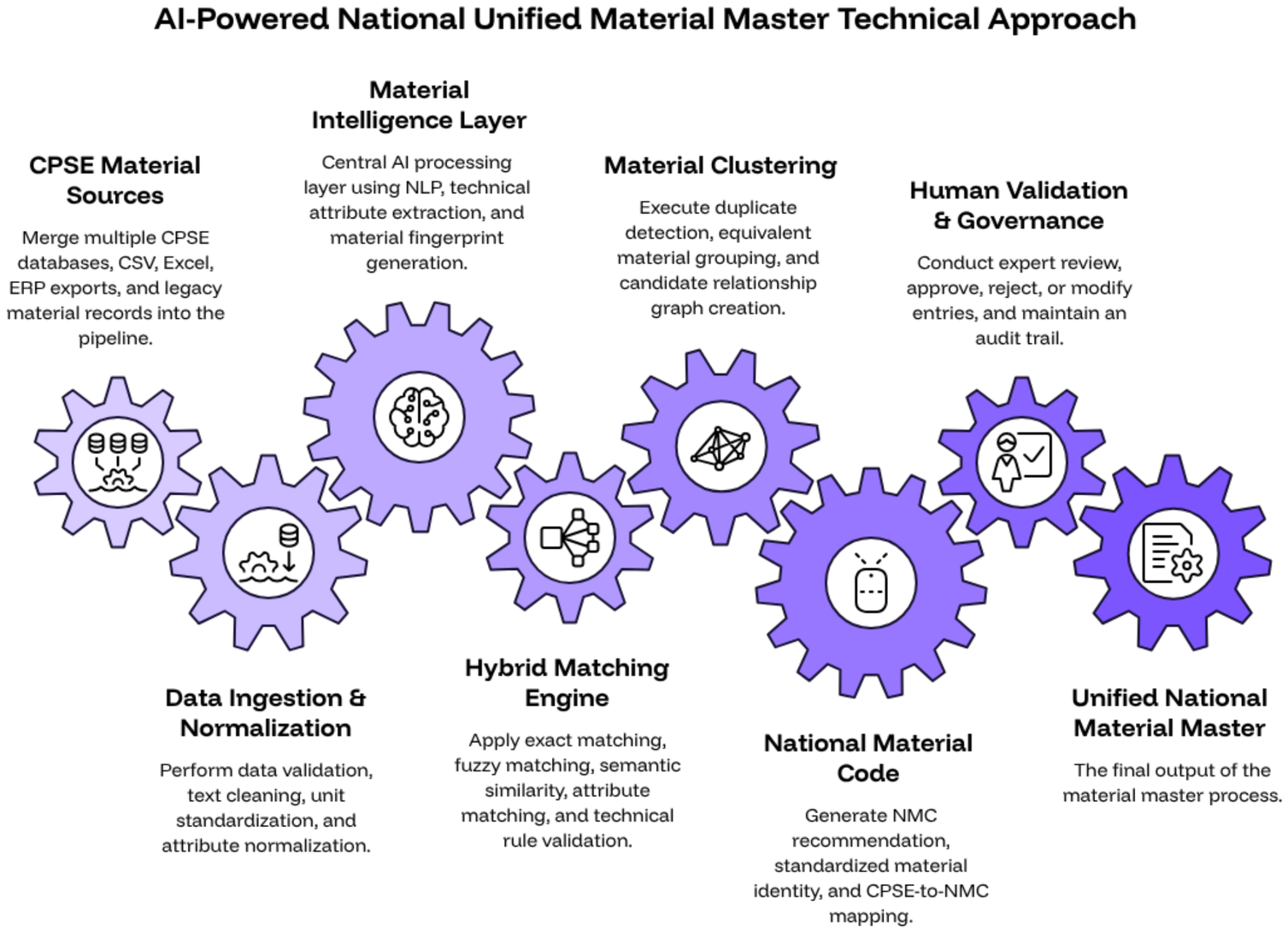
Backend: Python , FastAPI

AI / ML:

- NLP & semantic embeddings
- Entity / attribute extraction
- Fuzzy + semantic similarity
- ML-assisted duplicate detection
- LLM-assisted normalization

Data & Search:

- PostgreSQL
- Vector database / vector search
- Structured material maste
- Structured material master
- Audit & mapping store
- Infrastructure:
- REST APIs
- Containerized deployment
- Role-based access
- Scalable processing pipeline



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FEASIBILITY AND VIABILITY

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Feasibility

Can we build it?

Proven Technologies
- Uses established AI, NLP, ML and database technologies

Supports Heterogeneous Data
- Works with different CPSE material records and formats

Modular Architecture
- Enables incremental implementation

Preserves Existing Codes
- Existing CPSE codes remain intact and traceable

Human Validation
- Ensures controlled and reliable adoption

Challenges & Risks

What can go wrong?

Data Heterogeneity
- Inconsistent names, units, descriptions and specifications

Incorrect AI Matches
- Similar-looking descriptions or different materials

AI Hallucination / Low Confidence
- Incorrect recommendations or classifications

Legacy & Incomplete Data
- Missing, inconsistent or outdated records

Traceability & Governance
- Need for auditability and explainability

Scalability
- Large-scale processing across multiple CPSEs

Our Strategies

How do we solve it?

Data Normalization
- Standardize names, units and attributes

Hybrid Matching Engine
- Exact + Fuzzy + Semantic + Attribute + Rules

Confidence Scoring & Validation
- AI recommendations + human-in-the-loop

Data Quality Checks
- Handle missing/incomplete data and exceptions

CPSE ↔ NMC Mapping
- Maintain traceability and audit trail

Modular & Scalable Architecture
- Independent services and scalable processing

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IMPACT AND BENEFITS

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IMPACT & BENEFITS

For Government / CPSEs

- Unified view of materials across organizations
- Improved material master-data governance
- Better visibility into duplicate / equivalent records
- Consistent material identification across procurement ecosystems

For Procurement

- Improved material discovery and classification
- Reduced ambiguity in material descriptions
- Better identification of equivalent materials
- Foundation for more consistent procurement analytics

For Data & Technology

- National-scale standardized material identity
- Machine-readable material fingerprints
- Explainable AI-assisted mappings
- Complete CPSE ↔ National Code traceability

Strategic Impact

Fragmented Data → Standardized Identity → Interoperable Material Master → Better Data-Driven Procurement

Long-Term Vision

ONE NATION
ONE MATERIAL MASTER
ONE TRUSTED MATERIAL IDENTITY

Key Differentiators

- **AI-assisted** — Material intelligence
- **Explainable** — Evidence behind matches
- **Governed** — Human approval before authoritative mapping
- **Traceable** — Original CPSE identities preserved
- **Scalable** — Designed for multi-organization adoption

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RESEARCH AND REFERENCES

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- **ISO 8000-110:2021 — Data Quality: Master Data**
  Supports standardized exchange and semantic representation of master data across organizations.
- **ISO 8000-115:2024 — Master Data Quality Identifiers**
  Relevant to unique identification, semantic resolution and mapping of material identities.
- **Reimers & Gurevych (2019) — Sentence-BERT**
  Foundation for semantic similarity and embe
- **W3C PROV — Provenance Data Model**
  Relevant to maintaining traceability, provenance and audit history of standardized material mappings.
- **ISO/TS 8000-82:2022 — Data Quality Rules**
  Supports the use of machine-processable data rules for maintaining data integrity and reliability.

Github Link : https://github.com/NireekshaAP07/GOV